

Product Guide

MX-0812-SCL

Advanced 4K HDMI Matrix Switcher



WyreStorm®

**Fast Seamless Matrix with
Video Wall Support**

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Product Overview



Advanced 4K HDMI Fast Seamless Matrix

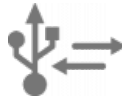
The “MX-0812-SCL” is a 4K fast seamless matrix, perfect for auditoriums, lecture halls, and large classrooms. Featuring 8 HDMI inputs and 12 HDMI outputs, it supports up to 4K resolution and enables dynamic splicing video walls. With advanced KVM functionality, it’s ideal for hybrid teaching and meetings. Seamless integration with touch panels or third-party controllers ensures effortless operation across diverse applications. Additionally, it supports Dante transmission, delivering superior audio quality for an enhanced experience.

Main Features



Reliable and High-Quality Resolution

Deliver sharp and clear 4K video for a fully immersive presentation experience.



Fast Seamless Switching

Eliminate black screens when switching between sources like laptops, PCs, and document cameras.



Video wall Feature Supporting

- Create video walls with up to 12 screens in layouts like 2x2, 3x2, and 3x3.
- Support multiple simultaneous video walls (e.g., 3 x [2x2]) and adjust bezels for improved image alignment and edge quality.



KVM for Hybrid Teaching & Meetings

Simplify cable management and streamline workflows with USB 3.0 ports for cameras, speakerphones, and keyboards. Designed for hybrid meetings and education, it enables seamless device switching through a single hub.



Built-in Dante(4x4) for Audio Transmission

Supporting Dante (4x4) to transmit audio signal, working directly with Dante Audio System, such as Dante audio DSP, Dante Mic, and Dante amplifiers.



Balanced Audio De-embedding

Integrate seamlessly with external audio systems for enhanced sound quality. Pair with a professional amplifier, speakers, and microphones to deliver an immersive audio experience for presenters and participants.



Versatile and Flexible Ethernet

Enhance hybrid meetings and education with reliable networking. Ethernet Bridge and Isolation ensure stability and security.



Centralized Control and Automation

Enable remote monitoring and control via Web-UI, RS232, and Telnet. Access real-time system status, configuration, and performance effortlessly.

Why Choose Us: 1. Immersive Viewing Experience and Video Wall

Deliver an immersive viewing experience with 4K@60 resolution, seamless switching, and video wall capabilities, ensuring stunning detail and captivating visuals for the audience.

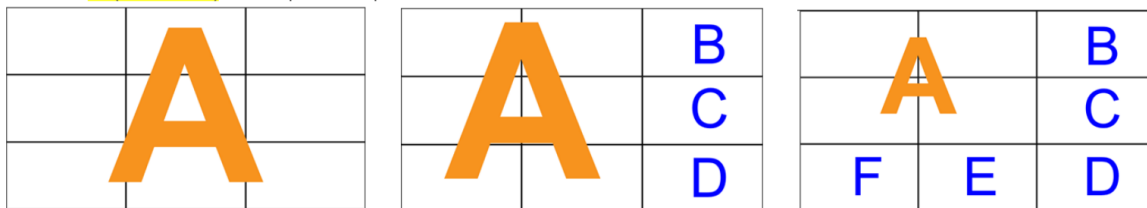
High-Quality Video Performance

Provide high-quality 4K resolution for detailed information on large-format screens, coupled with fast, seamless switching and no black screen to enhance the viewing experience.

Videowall 2x6(row x column)



Videowall 3x3(row x column) + 3x independent outputs



Multiple Splicing Layout

Multiple splicing layouts supporting, creating common layouts such as 2x2, 3x2, 4x2, 5x2, 6x2, 3x3, 3x4. And also creating multiple video walls simultaneously (3 x [2x2]), which allows for selecting different input sources.

Picture Adjustment

Bezel adjustment help guests to enhance viewing experience with stunning details.

Why Choose Us: 2. Creating Hybrid Working/Teaching Environment

Ensure effortless connectivity, smooth audio performance, and efficient operation in video conferencing, allowing all meeting participants to easily contribute and collaborate productively.

Effortless Connectivity

Works flawlessly with cameras, speakerphones, and other USB 3.0 peripherals, delivering a seamless and superior hybrid working experience.

Efficient Operation in Hybrid

Enhance the efficiency of hybrid meetings with a user-friendly Web-UI or by integrating with SYN-TOUCH10.

Smooth Audio Performance

Ensure clear and reliable audio-visual communication by directly working with Dante devices, essential for effective collaboration and decision-making.



Why Choose Us: 3. Effortless Device Management

Simplify setup with various control methods, making it easy for guests to connect and present without hassle.

Flexible Network Settings

Provide Ethernet isolation with 2x Ethernet ports for enhanced security and network stability, ideal for separating internal and guest networks.

Centralized Control

Manage and monitor all connected devices through an intuitive web interface or all-in-one touchpad controller, simplifying the administration process.

Remote Configuration

Configure the system remotely via SYGMA cloud device management, ensuring that everything runs smoothly without needing to be physically present.



Interface of the Product

Front Panel

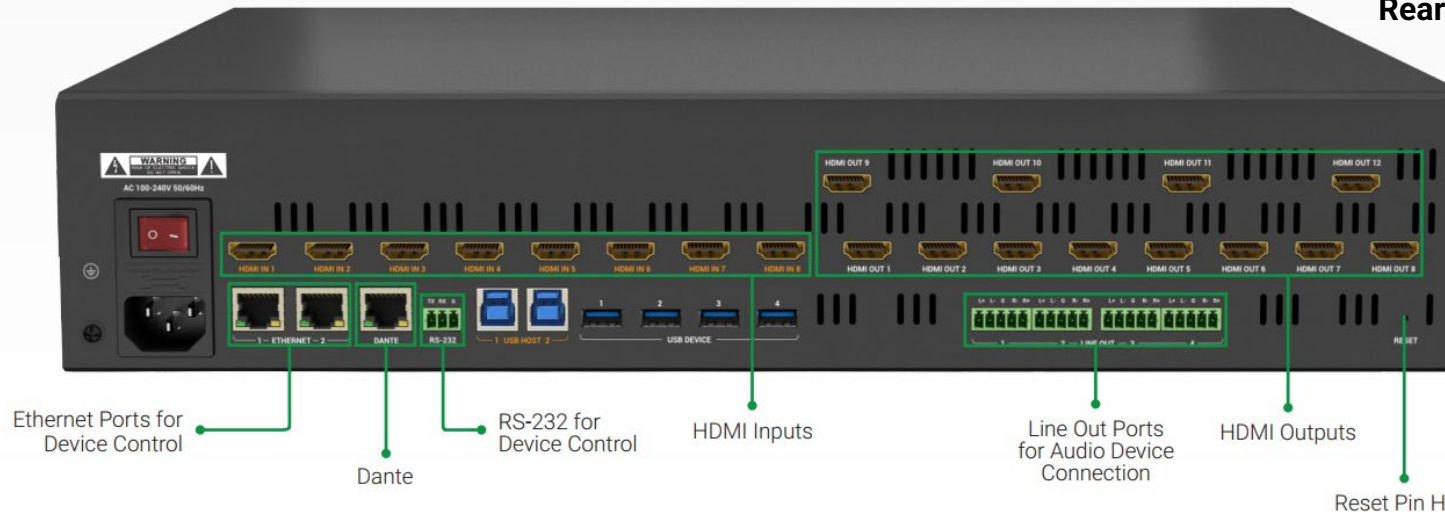


LCD Indicator

INFO, SWITCH, VIDEO
AUDIO Button

ENTER, CANCEL Button

Rear Panel



Ethernet Ports for
Device Control

Dante

RS-232 for
Device Control

HDMI Inputs

Line Out Ports
for Audio Device
Connection

HDMI Outputs

Reset Pin Hole

Featuring:

HDMI Inputs & Outputs:

Connect to HDMI sources and displays, supporting high-quality resolution video up to 4K@60Hz.

USB Hosts & Devices:

Connect to Web camera, speaker phone, Keyboard and mouse, very useful to share classroom signal with remote students.

3x Ethernet Ports:

One Ethernet port is Dante for audio transmission; Others ensure secure and stable network access for guests using their PCs, while allowing easy configuration of the system via a web interface.

Specifications & Packing List

Main Specifications

Input Video Port	8x HDMI
Output Video Port	12x HDMI
Output Video Signal	4K@60Hz 4:4:4 8bit, HDCP 2.3
Input/Output Resolutions Supported	VESA: 800x600⁸, 1024x768⁸, 1280x768⁸, 1280x800⁸, 1280x960⁸, 1280x1024⁸, 1360x768⁸, 1366x768⁸, 1440x900⁸, 1600x900⁸, 1600x1200⁸, 1680x1050⁸, 1920x1200⁸ SMPTE: 720x576P⁶, 1280x720P^{6,7,8}, 1920x1080P^{2,5,6,7,8}, 3840x2160^{2,3,5,6,8}, 4096x2160^{2,3,5,6,8} (4:2:0 8bit only) 2 = at 24 Hz, 3 = at 25 Hz, 5 = at 30 Hz, 6 = at 50 Hz, 7 = at 59.94 Hz, 8 = 60 Hz
Audio Format	HDMI IN/OUT/LINE OUT: PCM 2.0
Control Method	Front panel buttons, RS232, LAN (Telnet API & Web UI)
Power Supply	AC 100~240V 50/60Hz
Power Consumption (Max)	120W
Dimensions (WxHxD)	440mm x 88mm x 362mm/17.32" x 3.46" x 14.25"

What's in the Box?

- ☒ 1x MX-0812-SCL Matrix
- ☒ 1x AC Power Cord (US/UK/EU/AU)
- ☒ 1x 3.5mm 3-Pin Phoenix Male Connector
- ☒ 4x 3.5mm 5-Pin Phoenix Male Connector
- ☒ 2x Mounting Brackets (with Screws)
- ☒ 1x Quick Start Guide

Typical Connection & Applications

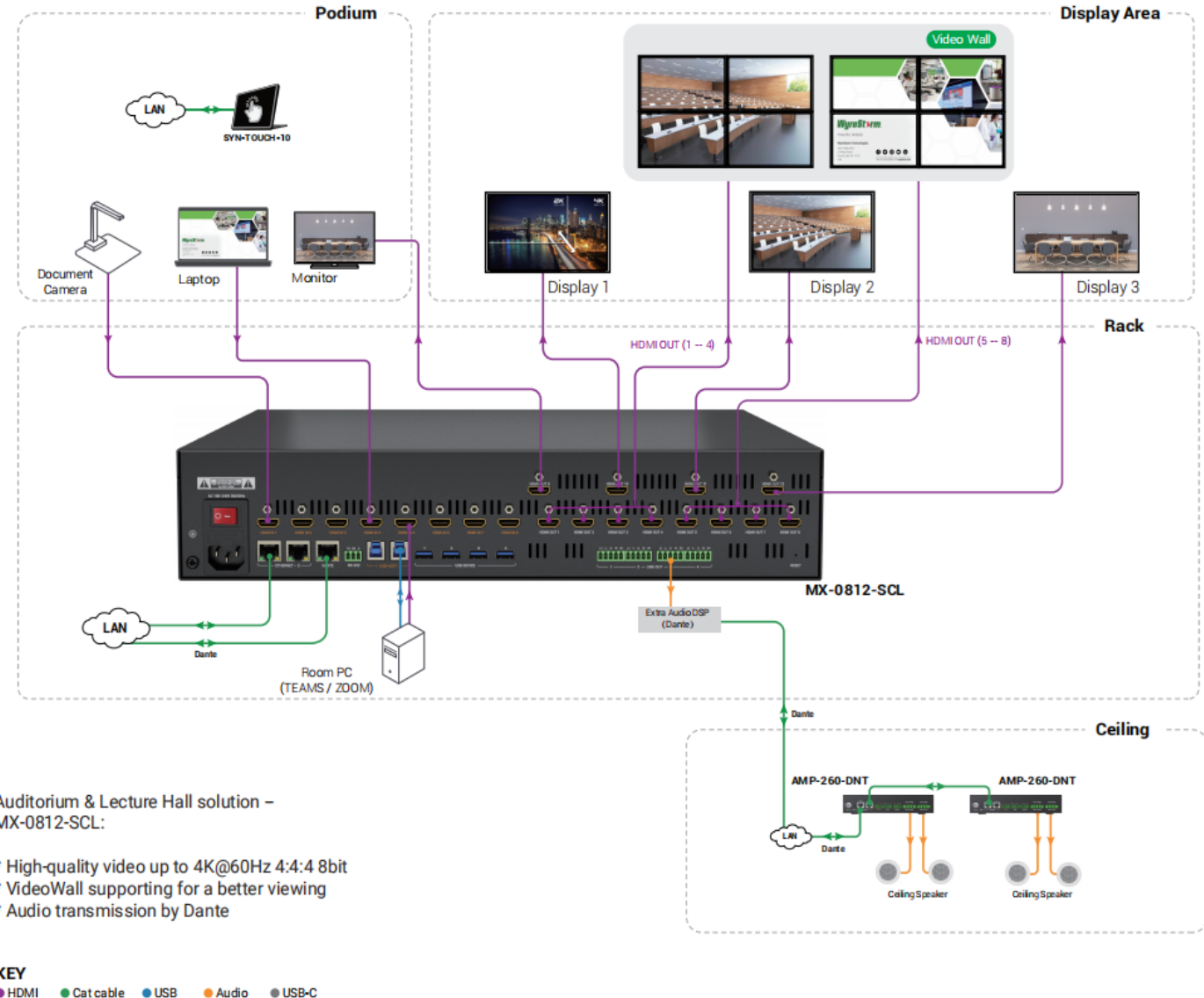
Application Scenario 1:

Auditorium & Lecture Hall (Video Wall)

This solution is ideal for auditoriums and lecture halls, connecting video sources to a video wall or displays with the MX-0812-SCL matrix, delivering clear visuals for students. Paired with Dante audio systems and speakers, it ensures high-quality sound to maintain focus. The intuitive touch panel simplifies presentation management, enhancing interactivity and engagement.

Key Benefits Include:

- **Multiple layout video wall:** Create multiple layout video walls, such as two 2x2 setups, to enhance presentation effects.
- **High-quality viewing experience:** Deliver a high-quality viewing experience with up to 4K@60Hz resolution and fast, seamless switching with no black screen.
- **Provide professional audio performance:** Provide professional audio performance by integrating with external audio DSPs, amplifiers, and ceiling speakers.
- **Dante (4x4):** Ensure seamless compatibility with other Dante-enabled devices through built-in Dante (4x4) technology.
- **Flexible control methods:** Simplify system control with flexible options, allowing professors and students to easily manage the entire system using a touch panel.



Typical Connection & Applications

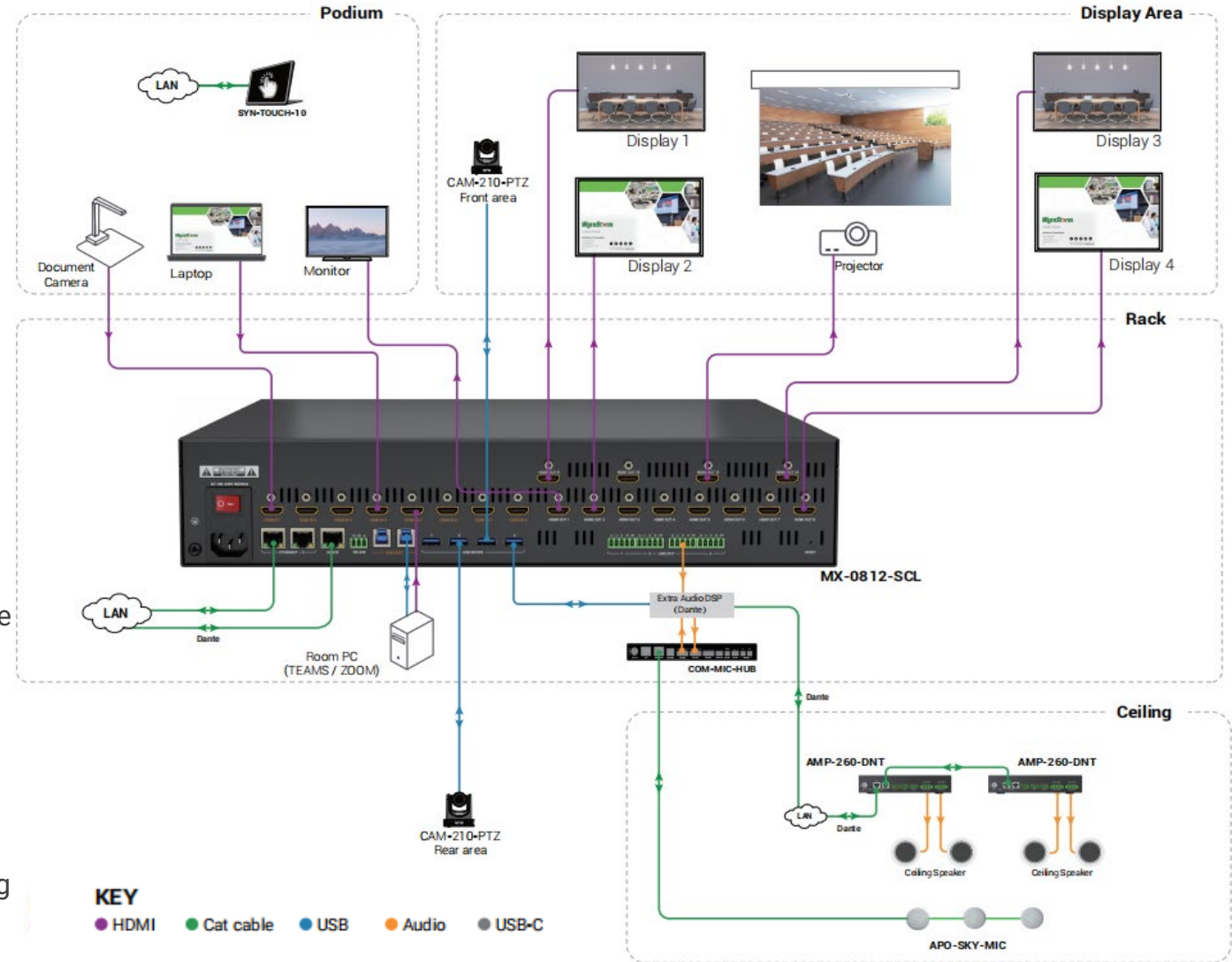
Application Scenario 2:

Auditorium & Lecture Hall (Hybrid)

This solution is ideal for hybrid learning in large spaces. The MX-0812-SCL matrix switch connects multiple video sources (laptops, document cameras, in-room PCs) to displays or projectors, delivering clear, vibrant 4K visuals for in-room students. Integrated USB support ensures seamless connection for remote students, providing them with high-quality audio and video. The intuitive touch control panel makes it easy for users to manage the system.

Key Benefits Include:

- **High-quality viewing experience:** Deliver up to 4K@60Hz resolution with fast, seamless switching and no black screen.
- **Provide professional audio performance:** Integrate with external audio DSPs, amplifiers, microphones, and ceiling speakers to enhance the audio experience.
- **Hybrid teaching & meeting:** Connect to camera, audio system, keyboard and mouse, we offer professional hybrid experience.
- **Dante (4x4):** Built-in Dante technology to ensure seamless compatibility with other Dante-enabled devices.
- **Flexible control methods:** Enable control via touch panel, allowing professors and students to easily manage the entire system.



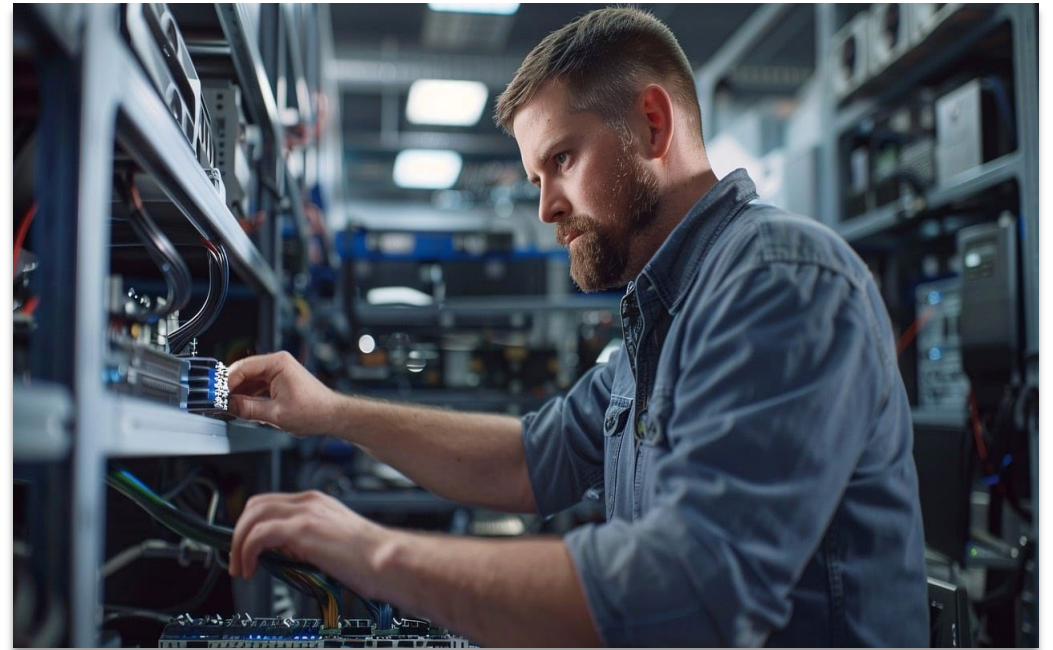
Comparison with Other Competitors

Features	Wyrestorm (MX-0812-SCL)	Kramer (VS-88H2A)	Atlona (AT-HDR-H2H-88MA)	Aten (VM5808HA)
HDMI	4K@60 444 8bit	4K@60 444 8bit	4K@60 420 10bit	4K@30
HDCP	HDCP 2.3	HDCP2.2	HDCP2.2	HDCP 1.4
Video Wall	Yes, M*N videowall layouts (M,N<=6 and M*N<=12)	No	No	Yes
KVM/USB	KVM Supports/USB3.0 + USB2.0	No	No	No
Audio Input	Dante	Audio IN	No	No
Audio Output	LINE OUT , Dante OUT	Audio OUT	Audio OUT	No
Audio DSP	No	Yes	No	No
Audio Matrix	12 x 8	24 x 24	8 x 8	8 x 8
ARC	No	Yes	No	No
Dante	Dante 4*4	No	No	No
Seamless Switching	Yes	Yes	Yes	Yes
Scaler	Yes	Yes	No	Yes
RS232	Yes	Yes	Yes	Yes
IR	No	No	Yes	No
Ethernet	2xLAN	1xLAN	1xLAN	1xLAN

Quick Installation Guide

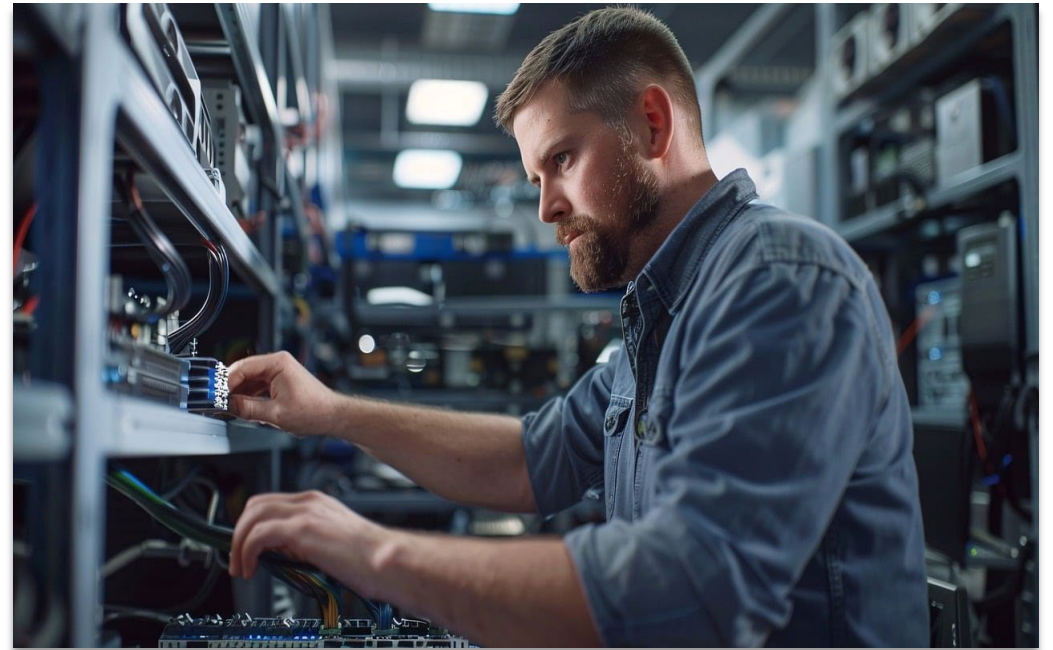
Effortlessly set up your fast seamless matrix with simple steps, ensuring seamless configuration and an exceptional viewing experience.

- **Connect Interfaces:** Follow the standard product connection guidelines and customize connections based on your environment's layout for optimal performance.
- **Start the Device:** Use SmartGUI to locate the device's IP address, then log into the Web-UI for easy equipment configuration.
- **Create Your Video Wall:** Create your layout using the Web-UI. For detailed instructions, refer to the user manual.
- **Enjoy Superior Visuals:** Experience stunning video clarity and immersive visuals tailored to your needs.



Important Notes for Installation

- Ensure all cables are securely connected to avoid performance issues.
- Check your network settings and confirm that the device is on the correct subnet.
- Double-check configuration settings in the Web-UI to optimize display and functionality.
- For complex installations, refer to the documentation or seek professional assistance if necessary.



Introduction of Web Configuration Interface

The Web UI of the MX-0812-SCL fast seamless matrix is designed for intuitive device management and control through any web browser. With the Web UI, you can easily configure and customize your videowall setup with the following features:

Web UI Functionality:

- **Interface:** Effortlessly manage device settings via a web browser.
- **Setting up:** Video routing, audio control, and USB settings or design custom video wall layouts.
- **Control:** Set up and adjust video inputs, outputs, and EDID profiles.

Key Web UI Configurations:

- **General Settings:** Configure IP settings, manage security, and perform firmware updates.
- **Video Inputs/Outputs:** Manage HDMI sources, HDCP compliance, and EDID profiles.
- **Video wall Settings:** Easy setting up to create the custom video wall layout.
- **Control Settings:** Enable RS-232 for device control and CEC for display control.

Final Thoughts

Q & A Session

Got questions? Don't hesitate to ask away.

